

Lecture Seven

1 Implicit Differentiation

1.1 Implicit and Explicit Function

Consider the function $y = 3x^2 - 5$, the variable y is explicitly written as a function of x , and every function can be expressed in the form $y = f(x)$ [y is a function of x] is called **explicit function**. But, there are some functions however are only implied by an equation. For instance, the function $y = \frac{1}{x}$ is defined implicitly by the equation $xy = 1$.

If we were asked to find $\frac{dy}{dx}$ for this equation, we could begin by writing y explicitly as a function of x and differentiating.

$$\left\| \begin{array}{l} \text{Implicit Form} \\ \text{Explicit Form} \\ \text{Derivative} \end{array} \right\| \left\| \begin{array}{l} xy = 1 \\ y = \frac{1}{x} = x^{-1} \\ \frac{dy}{dx} = -x^{-2} = \frac{-1}{x^2} \end{array} \right\|$$

This strategy works whenever we can solve for the function explicitly.

We cannot use this procedure when we are unable to solve for y as a function of x . For example, for the function $x^2 - 2y^2 + 4y = 2$, here it is very difficult to express y as a function of x explicitly. For this reason we will use **implicit differentiation**.

1.2 Guidelines for Implicit Differentiation

1. Differentiate both sides of the given equation with respect to x .
2. Collect all terms involving $\frac{dy}{dx}$ on the left side of the equation and move all other terms to the right side of the equation.
3. Factor $\frac{dy}{dx}$ out of the left side of the equation.
4. Solve for $\frac{dy}{dx}$ by dividing both sides of the equation by the left hand factor that does not contain $\frac{dy}{dx}$.

Example 1.2.1 Find $\frac{dy}{dx}$ for the given functions.

$$1. \quad y^3 + y^2 - 5y - x^2 = -4$$

Sol

Firstly Differentiate both sides of the equation w.r.t x

$$\frac{d}{dx}[y^3 + y^2 - 5y - x^2] = \frac{d}{dx}[-4]$$

we get

$$3y^2 \frac{dy}{dx} + 2y \frac{dy}{dx} - 5 \frac{dy}{dx} - 2x = 0$$

Secondly collect the $\frac{dy}{dx}$ terms on the left side of the equation.

$$3y^2 \frac{dy}{dx} + 2y \frac{dy}{dx} - 5 \frac{dy}{dx} = 2x$$

Thirdly Factor $\frac{dy}{dx}$ out of the left side of the equation

$$\frac{dy}{dx}(3y^2 + 2y - 5) = 2x$$

Fourthly solve for $\frac{dy}{dx}$ by dividing by $3y^2 + 2y - 5$, we get:

$$\frac{dy}{dx} = \frac{2x}{3y^2 + 2y - 5}$$

2. $3(x^2 + y^2)^2 = 100xy$ at the point $(3, 1)$.

Sol

$$\begin{aligned} \Rightarrow 6(x^2 + y^2) \left(2x + 2y \frac{dy}{dx} \right) &= 100 \left(x \frac{dy}{dx} + y \right) \\ \Rightarrow 12x(x^2 + y^2) + 12y(x^2 + y^2) \frac{dy}{dx} &= 100x \frac{dy}{dx} + 100y \\ \Rightarrow 12y(x^2 + y^2) \frac{dy}{dx} - 100x \frac{dy}{dx} &= 100y - 12x(x^2 + y^2) \\ \Rightarrow \frac{dy}{dx} [12y(x^2 + y^2) - 100x] &= 100y - 12x(x^2 + y^2) \\ \Rightarrow \frac{dy}{dx} = \frac{100y - 12x(x^2 + y^2)}{-100x + 12y(x^2 + y^2)} &= \frac{25y - 3x(x^2 + y^2)}{-25x + 3y(x^2 + y^2)} \\ \Rightarrow \frac{dy}{dx} \Big|_{(3,1)} = \frac{25(1) - 3(3)(3^2 + 1^2)}{-25(3) + 3(1)(3^2 + 1^2)} &= \frac{13}{9} \end{aligned}$$

2 Higher Derivatives

2.1 Second Derivative

We know the first derivative as:

$$\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Which we obtain it by differentiating the function $y = f(x)$ or implicitly by $f(x, y) = 0$. Now we can obtain **the second derivative** by differentiating the first derivative as follows:

$$\frac{d}{dx} \left[\frac{dy}{dx} \right] = \lim_{h \rightarrow 0} \frac{f'(x+h) - f'(x)}{h}$$

And we denote it by y'' , $\frac{d^2y}{dx^2}$, $f''(x)$, $\frac{d^2}{dx^2}[f(x)]$ and also $D_x^2[y]$.

Also, we can find the third derivative as same as the second, by differentiating the second derivative as follows

$$\frac{d}{dx} \left[\frac{d^2y}{dx^2} \right] = \lim_{h \rightarrow 0} \frac{f''(x+h) - f''(x)}{h}$$

and we denote it by y''' , $\frac{d^3 y}{dx^3}$, $f'''(x)$, $\frac{d^3}{dx^3}[f(x)]$ and also $D_x^3[y]$.

In general

$$\boxed{\frac{d^n y}{dx^n} = \frac{d}{dx} \left[\frac{d^{n-1} y}{dx^{n-1}} \right] = \lim_{h \rightarrow 0} \frac{f^{(n-1)}(x+h) - f^{(n-1)}(x)}{h}}$$

Example 2.1.2 1. $y = \sin 3x$, prove that $y^{(4)} = 81y$

Proof

$$\begin{aligned} y' &= 3 \cos 3x \\ y'' &= -9 \sin 3x \\ y''' &= -27 \cos 3x \\ y^{(4)} &= 81 \sin 3x = 81y \end{aligned}$$

2. $y = x^n$, prove that $y^{(n)} = n!$ and find the fifth derivative for $y = x^5$.

Sol

$$y = x^n \Rightarrow y' = nx^{n-1} \Rightarrow y'' = n(n-1)x^{n-2} \Rightarrow y''' = n(n-1)(n-2)x^{n-3}$$

This implies that

$$y^{(n)} = n(n-1)(n-2) \dots 3 \times 2 \times 1 \times x^{n-n} = n!$$

$$\text{Now for } y = x^5 \Rightarrow y^{(5)} = 5! = 120$$

3. Find $\frac{d^2 y}{dx^2}$ if $x^2 + xy + y^2 = a^2$.

Sol

Differentiate both sides w.r.t x

$$2x + (xy' + y) + 2yy' = 0 \quad (2.1.1)$$

Then collect all y' terms on the left side and then solve for y'

$$\begin{aligned} \Rightarrow xy' + 2yy' &= -2x - y \\ \Rightarrow y'(x + 2y) &= -(2x + y) \Rightarrow \boxed{y' = \frac{-(2x + y)}{x + 2y}} \end{aligned}$$

Now, by differentiating both sides of equation 2.1.1 w.r.t x, we get:

$$\begin{aligned} \Rightarrow 2 + xy'' + y' + y' + 2(yy' + (y')^2) &= 0 \\ \Rightarrow xy'' + 2yy'' &= -2 - 2y' - 2(y')^2 \\ \Rightarrow y''(x + 2y) &= -(2 + 2y' + 2(y')^2) \Rightarrow \boxed{y'' = \frac{-(2 + 2y' + 2(y')^2)}{x + 2y}} \end{aligned}$$

Then by substituting y' on equation of y'' , we get:

$$\begin{aligned}
 y'' &= - \left[2 + 2 \frac{-(2x+y)}{2y+x} + 2 \left(\frac{-(2x+y)}{2y+x} \right)^2 \right] \div (2y+x) \\
 &= - \left[\frac{2(2y+x)^2 - 2(2y+x)(2x+y) + 2(2x+y)^2}{(2y+x)^2} \right] \div (2y+x) \\
 &= - \left[\frac{2(4y^2 + 4xy + x^2) - 8xy - 4y^2 - 4x^2 - 2xy + 2(4x^2 + 4xy + y^2)}{(2y+x)^3} \right] \\
 &= \frac{-6(x^2 + xy + y^2)}{(x+2y)^3} = \frac{-6a^2}{(x+2y)^3}
 \end{aligned}$$

Exercise 2.1.1 For the function $x^2 + y^2 = a^2$

1. Prove that $y'' = \frac{-a^2}{y^3}$
2. Prove that $y''' = \frac{3a^2x}{y^5}$

2.2 Leibniz's Rule (Generalized Product Rule)

It gives us a formula to compute the n th derivative of the product of two functions. It extends the simple product rule to higher-order derivatives.

Theorem 2.2.1 If $f(x)$ and $g(x)$ are differentiable functions n -times, then n th derivative of their product is:

$$\frac{d}{dx} [f(x) \cdot g(x)] = \sum_{k=0}^n \binom{n}{k} f^{(n-k)}(x) \cdot g^{(k)}(x)$$

Where $\binom{n}{k}$ is the binomial coefficient (n choose k)
 $f^{(n-k)}(x)$ means the $(n-k)$ -th derivative of $f(x)$
 $g^{(k)}$ means the k -th derivative of $g(x)$

Note that, for $n = 1$ the formula reduces to the ordinary product rule.

$$(fg)' = f'(x)g(x) + f(x)g'(x)$$

Example 2.2.3 Find the fourth derivative for the function $y = x^4 \sin x$

Sol

$$\begin{aligned}
 \frac{d^4}{dx^4} [x^4 \sin x] &= \sum_{k=0}^4 \binom{4}{k} (x^4)^{(4-k)} \cdot (\sin x)^{(k)} = \binom{4}{0} (x^4)^{(4-0)} \cdot (\sin x)^{(0)} + \binom{4}{1} (x^4)^{(4-1)} \cdot (\sin x)^{(1)} \\
 &+ \binom{4}{2} (x^4)^{(4-2)} \cdot (\sin x)^{(2)} + \binom{4}{3} (x^4)^{(4-3)} \cdot (\sin x)^{(3)} + \binom{4}{4} (x^4)^{(4-4)} \cdot (\sin x)^{(4)} \\
 &= \binom{4}{0} (x^4)^{(4)} \cdot (\sin x) + \binom{4}{1} (x^4)^{(3)} \cdot (\sin x)^{(1)} + \binom{4}{2} (x^4)^{(2)} \cdot (\sin x)^{(2)} \\
 &+ \binom{4}{3} (x^4)^{(1)} \cdot (\sin x)^{(3)} + \binom{4}{4} (x^4)^{(0)} \cdot (\sin x)^{(4)} \\
 &= 4! \sin x + 4(4 \times 3 \times 2)x \cos x + 6(4 \times 3)x^2(-\sin x) + 4(4x^3)(-\cos x) + x^4 \sin x \\
 &= x^4 \sin x - 16x^3 \cos x - 72x^2 \sin x + 96x \cos x + 24 \sin x
 \end{aligned}$$

Exercise 2.2.2 Find $y^{(4)}$ if $y = x^4 e^{2x}$